Fig shows the structure of copper atom.It has 29 electrons which are arranged in different orbits as follows.

1 orbit = $2 \times 1^2 = 2$
2 orbit = $2 \times 2^2 = 8$
3 orbit = $2 \times 3^2 = 18$
4 orbit will have 1 electron

The atomic structure of all known elements can be shown in this way.

## 10.2 RECTIFIERS - TYPES

It has already been discussed in the previous chapter that a pn-junction conducts current easily when forward bias and practically no current flows when it is reverse biased.

"A pn-junction is known as a semiconductor or crystal diode"

For reasons associated with economics of generation and transmission, the electric power available is usually an a.c. supply. The supply voltage varies sinusoidally and has a frequency of 50 Hz. It is used for lighting, heating and also in electric motors.

Mostly all electronic devices require d.c. power for this proper operation. DC batteries are used for vehicles and rarely in commercial appliances. They are costly and require frequent charging or replacement. So we can get d.c. power from, a.c. power by using regulated d.c. power supply. It consists of transformer, rectifier filter and regulator.

### Classification of Rectifiers

The unidirectional characteristic active element ie., diode is used for this purpose. The rectifier convert an AC signal into DC signal. Rectifiers are classified into two types namely (i) Half wave rectifier (ii) Full wave rectifier. They are explained as below

### 10.2 (i) Half wave Rectifier

In half-wave rectification, the rectifier conducts current only during the positive half-cycle of input ac supply. The negative half-cycle of a.c. supply are suppressed ie., during negative half-cycle, no current is conducted and hence no voltage appears across the load.

#### Circuit details

Fig 10.2 shows the circuit where a single crystal diode acts as a half-wave rectifier. The a.c. supply is applied in series with the diode and load resistance $R_L$ through a transformer.

#### Operation

The a.c. voltage across the secondary winding AB changes polarities after every half-cycle. During the positive half-cycle of input a.c. voltage, end 'A' becomes positive with respect to 'B'. This makes the diode forward biased and hence it conducts current. During the negative half cycle, end A is negative with respect to B. Under this condition, the diode is reverse biased and it conducts no current. Therefore, current flows through the diode during positive half-cycle of inputs ac voltage only. It is blocked during the



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